

Trusten Lehmann-Karp

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EDUCATION

Harvey Mudd College

B.S., Engineering, Spanish Minor

Dean's List

Claremont, CA

Aug. 2024 - May 2028

- Relevant Coursework: Digital Electronics and Computer Engineering, Multivariable Calculus, Linear Algebra, Engineering Design, Engineering Systems, Physics Mechanics, Probability and Statistics, Computer Science

EXPERIENCE

Machine Learning Engineer

New York, NY (Remote)

Priori Sport

Jul. 2025 - Present

- Building machine learning models to quantify on-field behavior for professional and collegiate soccer teams (MLS, NCAA) using spatiotemporal tracking and event data.
- Designing and evaluating advanced performance metrics including expected goals (xG, AUC .761), expected assists (xA, AUC .981), pressing effectiveness, and a continuous-possession model, using 3.2M+ events.
- Developing end-to-end Python pipelines for feature engineering, model training, validation, and visualization, creating interpretable performance metrics bridging raw data and match film for coaching workflows.
- Collaborating with analysts and engineers to debug models, validate against match film, and refine labeling logic.
- Transitioned from internship to part-time role based on sustained ownership of modeling and analytics deliverables.

Software Engineer Intern

New York, NY (Remote)

Astral

May 2025 - Jul. 2025

- Developed a modular Python SDK for AI agent prototyping, enabling rapid integration with multiple LLM backends.
- Designed evaluation framework benchmarking 8+ LLM orchestration tools, informing product roadmap.
- Built scalable web-scraping pipelines to automatically ingest and normalize documentation across 10+ AI platforms.

AI Engineering Intern

New York, NY

Neuroaide

Dec. 2024 - Jan. 2025

- Structured and cleaned neuropsychological assessment data to improve quality of AI-generated clinical reports.
- Optimized data workflows and database schemas, reducing report error rates and turnaround time.

SELECTED PROJECTS

AI-Powered Consultant Knowledge Base

Full-Stack AI application | Next.js, FastAPI, pgvector

- Independently built a full-stack AI knowledge platform for querying PDF documents using LLMs and vector search.
- Implemented semantic memory with pgvector + OpenAI embeddings, enabling project-level and global retrieval.
- Deployed production system on Vercel (frontend) and Railway (backend) with usage analytics.

Agentic AI College Advisor (P-ai Club)

Project Management | Full Stack Development

- Lead PM 7-member team building AI advisor using LLMs and vector search to generate personalized degree plans.
- Coordinated model iteration, evaluation, and frontend/backend integration to align outputs with real life uses.

Profile Matching Algorithm (P-ai Club)

Model Development | BERT Fine-tuning

- Fine-tuned BERT model for intelligent profile matching for mobile app, improving recommendation accuracy.
- Collaborated with product and design teams to iterate on model behavior using live feedback and engagement metrics.

SKILLS

Programming Languages: Python, Java, C

Machine Learning: Logistic Regression, Random Forest, XGBoost, BERT, Spatiotemporal Modeling, Model Evaluation

AI & LLM: LLM APIs, Prompt Engineering, Embeddings, Vector Databases, RAG, Agentic Workflows

Data & Backend: Data Pipelines, Event-Based Modeling, FastAPI, Supabase, REST APIs

Frameworks & Tools: scikit-learn, Pandas, NumPy, Next.js, Jupyter, Matplotlib, Git

Languages: Spanish (Intermediate)

ACTIVITIES

Pomona AI Club (P-ai)

Claremont, CA

Project Manager

Jan. 2025 - Present

- Leading and mentoring students in developing applied AI projects and industry-relevant technical skills.

Claremont-Mudd-Scripps Men's Soccer, NCAA DIII

Claremont, CA

Recruited Student-Athlete

Aug. 2024 - Present

- Developing valuable life and leadership skills through competition, 2024 SCIAC Champion, NCAA Sweet-16